

Circuit family $\times$ initialisation $\times$ dataset

A progressive block-size study on DIV2K and TU-Berlin ($m = n = 8, 256 \times 256$)

Generated 2026-05-25 · mean test PSNR over the test split

Summary. We train five quantum-circuit transform families — rich (complex $U(4)$ gates), qft, tebd, entangled_qft, mera — under a block-size curriculum, from two initialisations (identity-operator and random), on two datasets (natural images and sketches). Two themes emerge. **The dataset sets the regime:** on DIV2K the block-size curve is flat and rich leads, while on block-sparse TU-Berlin the curve inverts (small blocks win), the family ranking flips with the rate, and classical block-DCT is near-lossless. **And a common rate-crossover holds on both:** classical block-DCT dominates the learned circuits at light compression but the learned circuits overtake it as compression tightens. Throughout, rich (complex $U(4)$) leads the learned families, and **identity init is never worse than random** — often dramatically better.

1 Setup

Each *_progressive sweep trains a circuit family under a **block-size curriculum**: at stage k the inner basis is a $2^k \times 2^k$ circuit replicated across the image by a BlockedBasis wrapper (the bare full circuit at $k = 8$). Stages are independent — no warm-start — and each trains for 1008 steps (generalized preset, 112 epochs) under an MSELoss that keeps the top- k coefficients by magnitude.

Families	rich (complex $U(4)$ 2-qubit gates), qft, tebd, entangled_qft, mera
Initialisations	identity — every gate dropped to its manifold identity ($H \rightarrow I_2$, controlled-phase $\rightarrow$ phase 0), so the circuit starts as the identity operator. random — rich: Haar $U(2)/U(4)$; qft: Haar $U(2)$ on Hadamards + random controlled-phase angles; tebd/entangled_qft/mera: native seeded random gates.
Stage range	rich/qft: $k = 1..8$; tebd/entangled_qft: $k = 2..8$; mera: $k \in \{2, 4, 8\}$ (m must be a power of 2). Tables show $k = 2..8$.
Datasets	DIV2K-8q — 500 natural images, centre-crop + LANCZOS to 256×256 . TU-Berlin — 500 line-drawing sketches, $\approx 96.5\%$ white pixels, same preprocessing.
Metric	mean test PSNR @ keep-ratio ρ (fraction of coefficients retained); $\rho = 0.20$ is the headline rate.

2 DIV2K-8q: natural images

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	32.9	33.7	33.6	33.4	33.4	33.6	33.6
rich	random	32.9	33.4	33.5	32.9	33.2	32.9	33.4
qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
qft	random	32	32.3	31.7	30.9	30.8	31.3	31.3
tebd	identity	32	32.3	32	31.7	31.7	31.7	31.7
tebd	random	32	32.3	31.7	30.9	30.8	30.9	30.9
ent-qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
ent-qft	random	32	32.3	31.7	30.9	31.3	31.3	31.3
mera	identity	32	n/a	32	n/a	n/a	n/a	31.7
mera	random	32	n/a	31.7	n/a	n/a	n/a	30.9

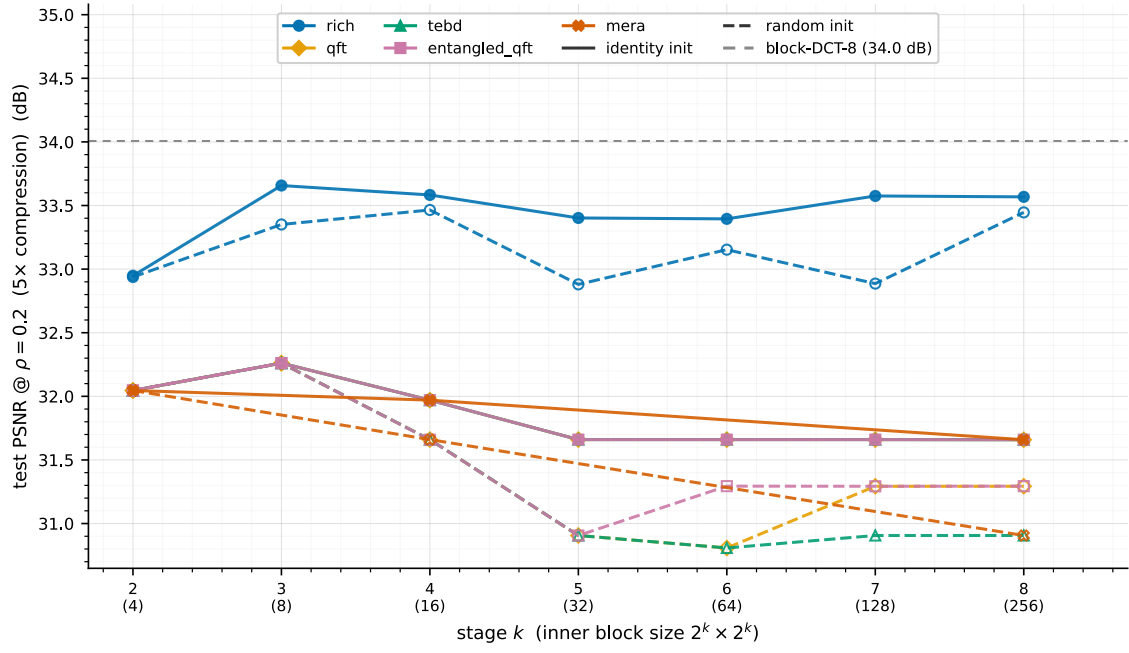


Figure 1: DIV2K, $\rho = 0.20$ (5 $\times$). One colour per family, solid = identity / dashed = random init; grey dashed = classical block-DCT-8 reference. rich leads the learned families but sits just below block-DCT.

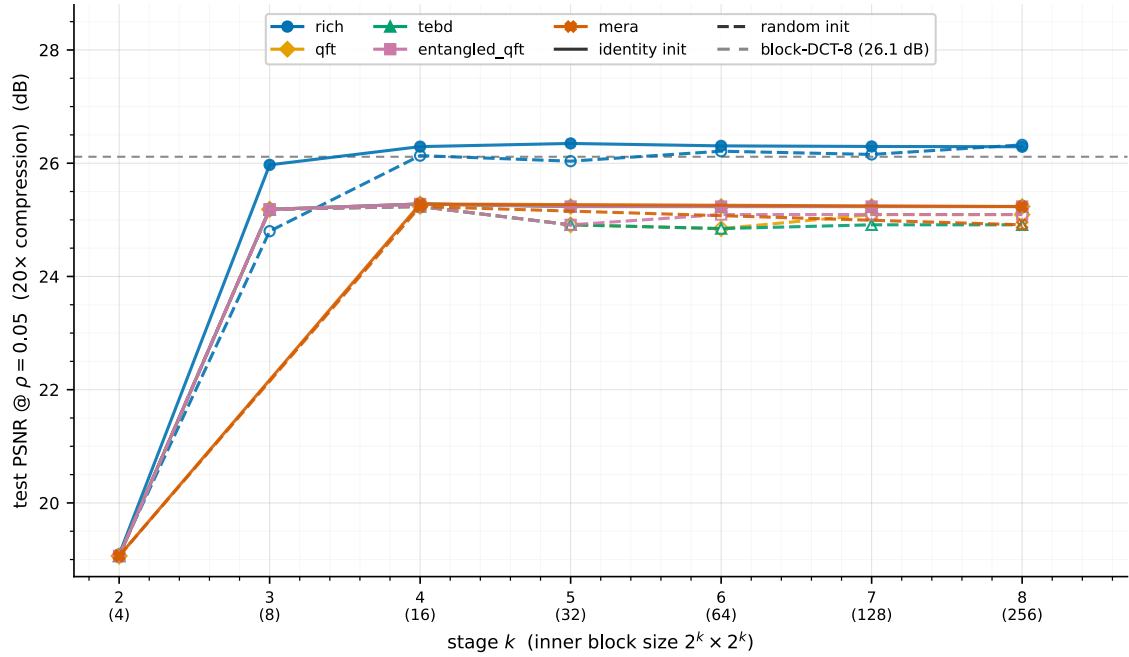


Figure 2: DIV2K, $\rho = 0.05$ (20 $\times$). The families bunch near block-DCT-8; rich draws level with it.

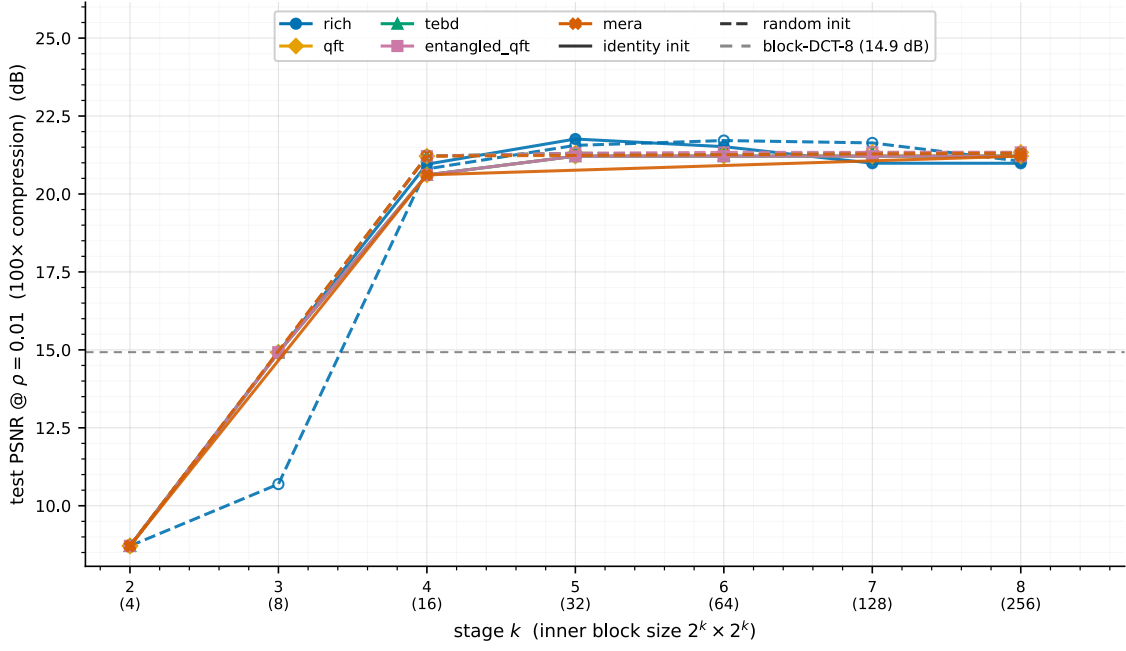


Figure 3: DIV2K, $\rho = 0.01$ (100 $\times$). Every learned family clears block-DCT-8 by ≈ 6 –7 dB – the learned transforms win at aggressive compression.

Findings.

- **rich leads the learned families but classical block-DCT wins at the light rate.** At $\rho = 0.20$ rich peaks at 33.7 dB, ≈ 1.5 dB above every other family – yet still just below the classical block-DCT-8 reference (34.0 dB). Complex $U(4)$ is the one architectural feature that helps, but it does not beat the fixed block transform here.
- **The learned circuits overtake block-DCT as compression tightens.** block-DCT-8 falls faster with the rate (34.0 $\rightarrow$ 26.1 $\rightarrow$ 14.9 dB at $\rho = 0.20/0.05/0.01$) than the learned circuits do: rich draws level at $\rho = 0.05$ (26.4 vs 26.1) and **every** family clears block-DCT by ≈ 6 –7 dB at $\rho = 0.01$ (≈ 21 vs 14.9).
- **Under identity init the four QFT-derived families are bit-identical** at every k despite very different gate counts ($k = 8$: qft 72, tebd 32, entangled_qft 80, mera 44). From an identity start they all converge to the same QFT operator – the extra gates train to no-ops.
- **Identity init $\geq$ random** for every family; the curve is essentially flat in k (≈ 33 dB for rich, ≈ 31.7 for the rest) at $\rho = 0.20$.

3 TU-Berlin: sketches (a different regime)

Sketches are **block-sparse** — most 8×8 tiles are blank — so a classical block transform is near-lossless: block-DCT-8 reaches 100.7 dB mean / 94.5 median at $\rho = 0.20$ (versus ≈ 32 on DIV2K), but only 36.5 dB at $\rho = 0.05$. We therefore report two compression rates.

3.1 Light compression: $\rho = 0.20$ (5 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	81.1	78.8	74.9	62.1	62.1	61.5	61.5
rich	random	79.5	73.8	62.6	48.8	41.5	33.9	44.9
qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
qft	random	87.1	83.1	65.6	48.5	39.6	43.2	44.2
tebd	identity	85	87.5	69.6	51.4	50.2	62.9	62.9
tebd	random	88.2	87.2	69.5	45.9	34.6	44.2	38.1
ent-qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
ent-qft	random	82.7	85.4	70.1	45.7	38.4	44.2	44.2
mera	identity	85	n/a	69.6	n/a	n/a	n/a	62.9
mera	random	88.2	n/a	69.4	n/a	n/a	n/a	38.4

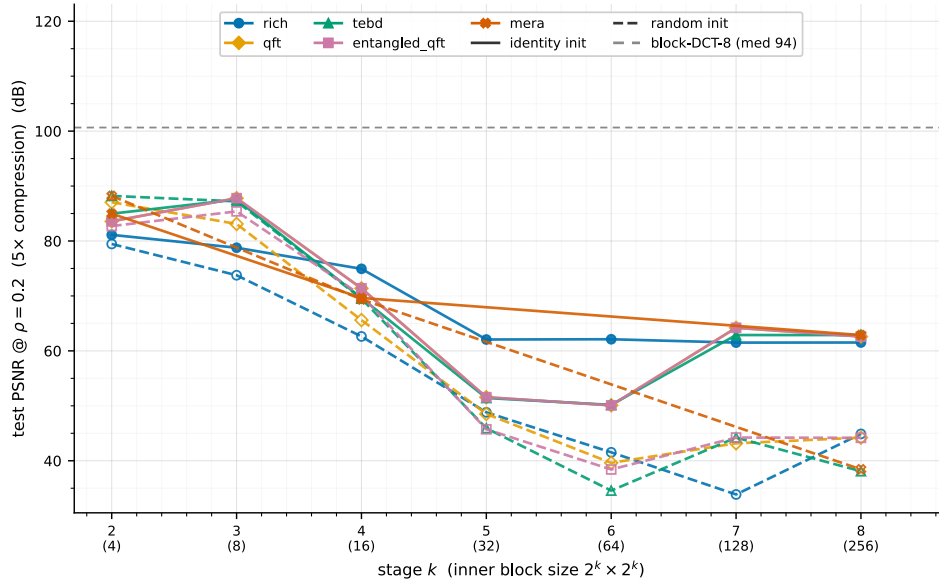


Figure 4: TU-Berlin, $\rho = 0.20$. The curve **falls** with block size; the QFT-derived families peak above rich at $k = 3$; all trail block-DCT-8 (grey).

3.2 Heavy compression: $\rho = 0.05$ (20 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	7.6	37.3	37.2	34.8	35.5	35.5	36.1
rich	random	7.6	30.6	31.6	29.6	28.5	25.7	29.3
qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
qft	random	7.6	29.8	31	28.8	27.3	28	28.4
tebd	identity	7.6	33.5	31.5	29.7	29.7	31.5	31.5
tebd	random	7.6	33.5	31.5	28.5	26.1	28.5	27.1
ent-qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
ent-qft	random	7.6	33.5	31.5	28.5	27.1	28.5	28.5
mera	identity	7.6	n/a	31.5	n/a	n/a	n/a	31.5
mera	random	7.6	n/a	31.5	n/a	n/a	n/a	27.1

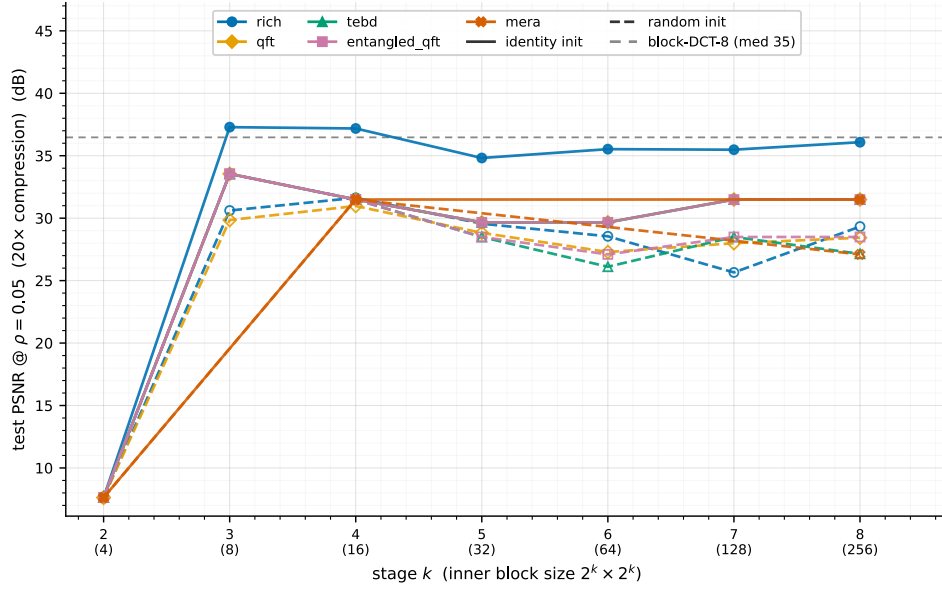


Figure 5: TU-Berlin, $\rho = 0.05$. rich (identity) leads and edges **above** block-DCT-8 at $k = 3, 4$; the per- k curve flattens.

3.3 Very heavy compression: $\rho = 0.01$ (100 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	0.9	4.8	4.8	22.1	21.7	22.2	22.9
rich	random	0.9	1.9	18.7	21.8	22.9	22.1	23
qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
qft	random	0.9	1.9	21.8	22.9	22.5	22.5	23.1
tebd	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
tebd	random	0.9	4.8	23.2	23.2	22.3	23.2	22.7
ent-qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
ent-qft	random	0.9	4.8	23.2	23.2	22.7	23.2	23.2
mera	identity	0.9	n/a	23.2	n/a	n/a	n/a	23.2
mera	random	0.9	n/a	23.2	n/a	n/a	n/a	22.7

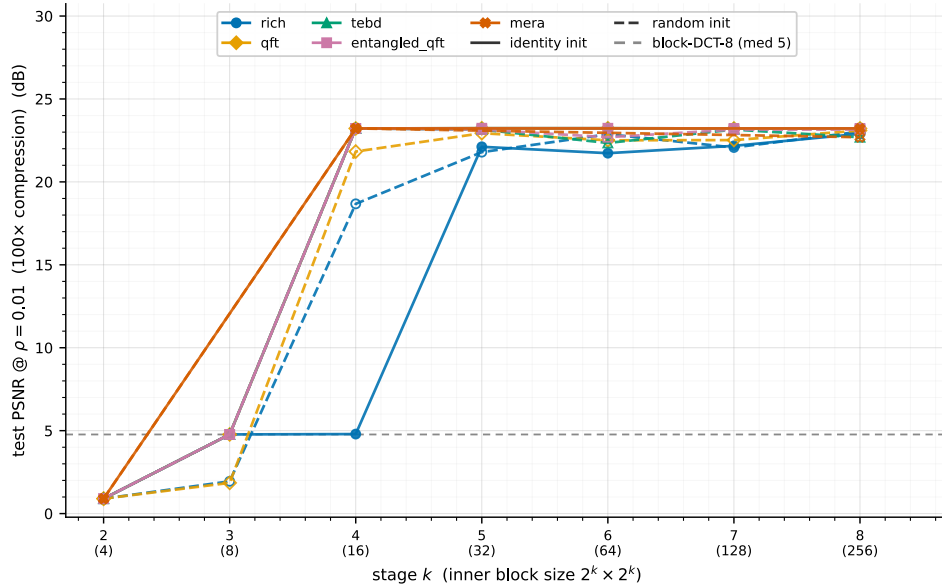


Figure 6: TU-Berlin, $\rho = 0.01$. block-DCT-8 **collapses** to ≈ 4.8 dB while the learned circuits plateau ≈ 23 dB — clearing it by ≈ 18 dB. An even sharper version of the heavy-compression crossover seen on DIV2K.

Findings.

- **The block-size curve inverts at the light rate.** Small blocks win — each blank tile costs almost no coefficients — the opposite of DIV2K’s flat curve. (At the heavy rates the curves instead rise then plateau in k .)
- **The family ranking flips with the rate.** At $\rho = 0.20$ the QFT-derived families peak ≈ 88 dB at $k = 3$, above rich (≈ 79); at $\rho = 0.05$ rich (identity) leads (≈ 37 dB); at $\rho = 0.01$ the QFT-derived families edge back ahead (≈ 23.2 vs rich 22.9).
- **The heavy-compression crossover is extreme here.** block-DCT-8 falls $100.7 \rightarrow 36.5 \rightarrow 4.8$ dB across $\rho = 0.20/0.05/0.01$: it is near-lossless at the light rate but the learned circuits beat it by ≈ 1 dB at $\rho = 0.05$ and by ≈ 18 dB at $\rho = 0.01$.
- **Identity $\triangleright$ random** — the gap reaches ≈ 25 – 30 dB at large k , far wider than on DIV2K. The QFT-family identity equivalence partly survives: $\text{qft} \equiv \text{entangled_qft}$ at all k , and all four QFT-derived families coincide at $\rho = 0.05$.

4 Cross-dataset discussion

	DIV2K (natural)	TU-Berlin (sketch)
block-size curve	flat in k	falls with k (small blocks win)
best learned family	rich (everywhere)	QFT-family @ $\rho=0.2$ & 0.01 ; rich @ $\rho=0.05$
vs classical block-DCT	block-DCT wins @ $\rho=0.2$; learned overtakes by $\rho=0.01$	block-DCT wins @ $\rho=0.2$; learned overtakes @ $\rho=0.05$ (+1 dB) & 0.01 (+18 dB)
identity vs random	identity $\geq$ random (< 1 dB)	identity $\triangleright$ random (up to 25–30 dB)
absolute PSNR @ $\rho=0.2$	≈ 31 – 34 dB	≈ 60 – 88 dB

The contrast is governed by **energy compaction relative to the data’s sparsity structure**. Natural-image energy is spread across scales, so a global / large circuit and the complex- $U(4)$ richness of rich help, and the curve is flat; sketches are block-sparse, so a small per-block transform already captures almost everything and a fixed block-DCT is near-lossless. On both datasets the fixed block transform degrades faster than the learned circuits as the rate tightens, so learning only pays off at aggressive compression.

5 Conclusions

1. **The dataset sets the regime.** Conclusions about “which circuit family is best” do not transfer between natural images and sketches; even the sign of the block-size trend flips.
2. **Learned circuits beat classical block-DCT only at aggressive compression.** At the light rate block-DCT wins on both datasets; the learned transforms overtake it as ρ shrinks (by $\rho = 0.01$ on DIV2K, $\rho = 0.05$ on sketches for rich).
3. **Complex $U(4)$ (rich) is the only architectural feature that consistently helps** — it leads the learned families on both datasets.
4. **Identity initialisation dominates random** everywhere, decisively so on sketches; the structured identity start is a strong, cheap prior.

Data: results/family_init_matrix/{div2k_8q,tuberlin_8q}/<family>_<init>/. Reproduce a sweep with experiments/qft_progressive.py -- dataset <d> -- family <f> --init <i>; regenerate figures with each report’s render_fig.py.